

Submission PHYS1-FINAL-SUB06

Question PHYS1-FINAL-Q01

notation changes halfway:

1. $v \approx v_i + at \approx 8 + 2(5) \approx 18 \text{ m/s}$

2. $\Delta x \approx v_i t + \frac{1}{2}at^2 \approx 8(5) + \frac{1}{2}(2)(5)^2 \approx 65 \text{ m}$

marks, subscripts, and units are mixed up

result written as: $v=18 \text{ m/s}$; displacement=65 m (please interpret)

Question PHYS1-FINAL-Q02

notation changes halfway:

1. $N \approx mg \approx 19.6 \text{ N}$; friction $\approx \mu N \approx 3.9 \text{ N}$

2. $\Sigma F \approx 13 - 3.9 \approx 9.1 \text{ N}$

3. $a \approx \Sigma F/m \approx 4.54 \text{ m/s}^2$

marks, subscripts, and units are mixed up

result written as: $a=4.54 \text{ m/s}^2$; friction opposes motion (please interpret)

Question PHYS1-FINAL-Q03

notation changes halfway:

1. $mgh \approx \frac{1}{2}mv^2 \Rightarrow v \approx \sqrt{2gh} \approx \sqrt{2 \cdot 9.8 \cdot 6}$

2. $v \approx 10.84 \text{ m/s}$; mass cancels

marks, subscripts, and units are mixed up

result written as: $v=10.84 \text{ m/s}$ (please interpret)

Question PHYS1-FINAL-Q04

notation changes halfway:

1. $p_{\text{total, before}} \approx p_{\text{total, after}}$

2. It applies when external impulse on the system is negligible.

marks, subscripts, and units are mixed up

result written as: total momentum before equals after when external impulse is negligible (please interpret)

Question PHYS1-FINAL-Q05

notation changes halfway:

1. Impulse is area under $F(t)$: $J \approx F\Delta t \approx 6(0.5) \approx 3 \text{ N}\cdot\text{s}$.

2. $\Delta p \approx J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

marks, subscripts, and units are mixed up

result written as: impulse=3 N·s; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (please interpret)

Question PHYS1-FINAL-Q06

notation changes halfway:

1. $v \approx v_i + at \approx 5 + 2(5) \approx 15 \text{ m/s}$

2. $\Delta x \approx v_i t + \frac{1}{2}at^2 \approx 5(5) + \frac{1}{2}(2)(5)^2 \approx 50 \text{ m}$

marks, subscripts, and units are mixed up

result written as: $v=15 \text{ m/s}$; displacement=50 m (please interpret)

Question PHYS1-FINAL-Q07

notation changes halfway:

1. $N \approx mg \approx 68.6 \text{ N}$; friction $\approx \mu N \approx 13.7 \text{ N}$
2. $\Sigma F \approx 23 - 13.7 \approx 9.3 \text{ N}$
3. $a \approx \Sigma F/m \approx 1.33 \text{ m/s}^2$

marks, subscripts, and units are mixed up

result written as: $a=1.33 \text{ m/s}^2$; friction opposes motion (please interpret)

Question PHYS1-FINAL-Q08

notation changes halfway:

1. $mgh \approx \frac{1}{2}mv^2 \Rightarrow v \approx \sqrt{(2gh)} \approx \sqrt{(2 \cdot 9.8 \cdot 5)}$
2. $v \approx 9.90 \text{ m/s}$; mass cancels

marks, subscripts, and units are mixed up

result written as: $v=9.90 \text{ m/s}$ (please interpret)

Question PHYS1-FINAL-Q09

notation changes halfway:

1. $p_{\text{total,before}} \approx p_{\text{total,after}}$
2. It applies when external impulse on the system is negligible.

marks, subscripts, and units are mixed up

result written as: total momentum before equals after when external impulse is negligible (please interpret)

Question PHYS1-FINAL-Q10

notation changes halfway:

1. Impulse is area under $F(t)$: $J \approx F\Delta t \approx 6(0.5) \approx 3 \text{ N}\cdot\text{s}$.
2. $\Delta p \approx J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

marks, subscripts, and units are mixed up

result written as: impulse= $3 \text{ N}\cdot\text{s}$; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (please interpret)